From the Gap Ratio to the Cabibbo Doublet and Beyond

One mismatch. One particle. Sixteen findings. The integers spoke.

From the gap ratio to the Cabibbo Doublet and beyond. 16 findings, 15 papers, 6 scripts, 98 checks, 1 particle, 0 contradictions.

Registry: [HOWL-DISC-11-2026]

Series Path: [HOWL-DISC-6-2026] → [HOWL-DISC-7-2026] → [HOWL-DISC-8-2026] → [HOWL-DISC-11-2026]

DOI: 10.5281/zenodo.19528556

Date: April 2 2026

Domain: Session Documentation, Capstone

Status: Complete

AI Usage Disclosure: Only the top metadata, figures, refs and final copyright sections were edited by the author. All paper content was LLM-generated using Anthropic's Claude Opus 4.6.

Abstract

Session 3 of the HOWL series began with a question — do the three SM gauge couplings unify? — and ended with a particle. The SM gap ratio 218/115 = 1.896 misses the measured 1.358 by 40%. The SM does not unify. Enumerating 15 single-multiplet extensions in exact Fraction arithmetic identified a unique minimal survivor: the Cabibbo Doublet (3,2,1/6), a vector-like quark doublet with gap ratio 38/27 = 1.407 and M_GUT = 10^15.5 at the proton decay boundary. Three independent experimental anomalies — CKM first-row unitarity deficit at 2.5-4 sigma, the LEP forward-backward b-quark asymmetry at ~3 sigma, and the Higgs signal strength excess at ~2 sigma — each independently point to the same (3,2,1/6) representation at 1.5-6 TeV. Two-loop corrections improve unification from Delta(1/alpha_3) = -1.17 (one-loop) to -0.40 (two-loop), with the residual within standard GUT threshold correction range. Session 3 produced 16 findings documented in 15 papers, verified 6 scripts with 98 total checks, identified 1 new particle, generated 3 experimental tests, closed 2 investigation paths, and extended the database to 146 entries with 38/38 checks. The integers identified a specific BSM particle. The universe has not yet confirmed it.

---

1. The Starting Point

Session 3 inherited from Sessions 1-2: the DATA-2 database (123 entries in Q335 = 2^335 integer rational basis), the mathematical framework (MATH-1 through MATH-5: R_2 = pi/4, R_4 = pi^2/32, Q335 basis, n-ball remainder sequence), and 11 physics papers (PHYS-1 through PHYS-11: mass as inertia, couplings run, confinement wall, vacuum permittivity, theta_QCD = 0, Koide decomposition, QED series, R_2 in 9/9 domains, electroweak anatomy). The parameter count stood at 17 (19 SM parameters minus theta_QCD derived in PHYS-7, minus m_tau conditional on Koide a^2 = 2 in PHYS-8). The operational rules (R.1-R.6) and writing rules (W.1-W.8) were established.

The question that launched Session 3: the beta coefficients b_1 = 41/10, b_2 = -19/6, b_3 = -7 are exact rationals from the gauge group. The coupling constants at M_Z are measured. Do the three lines 1/alpha_i(mu) meet at a point?

---

2. The Arc

The session ran in four phases.

Phase 1: Computation. Scripts built and verified before any paper was written. The GUT running script (sin2_theta_w_1.py, 9/9 checks) computed the SM gap ratio, enumerated 15 BSM candidates, and identified the Cabibbo Doublet. The A_2 decomposition script (a_2_decomposition_0.py, 7/7 checks) decomposed the QED two-loop coefficient into three pieces. The Bessel PSLQ script (bessel_pslq_0.py, 6/6 checks) extended the independence record from 72/72 to 82/82. The DATA-3 verification (data_2_to_3_test_1.py, 32/32 checks) confirmed all 123 database entries.

Phase 2: Discovery. 16 findings emerged from the computational results. The central finding: the Cabibbo Doublet (3,2,1/6) is the minimal single-multiplet extension that fixes the gap ratio, and three independent experimental anomalies converge on the same representation from a completely different direction.

Phase 3: Writing. One paper per finding, with supporting appendix tables. 12 PHYS papers (PHYS-12 through PHYS-23), 1 MATH paper (MATH-6), 1 DATA paper (DATA-4), and 1 database record (Cabibbo Doublet specification).

Phase 4: Two-loop extension. The unification script (unification_test.py, 6/6 checks) computed the two-loop correction, reducing the unification miss from -1.17 to -0.40. Supporting tables written for the pending PHYS-24 paper.

---

3. The Sixteen Findings

#	Finding	Paper	Content	Status
1	Electroweak integer anatomy	PHYS-12	7 inputs, 11 outputs, all integers from gauge group	Published

#	Finding	Paper	Content	Status
2	SM does not unify	PHYS-13	Gap ratio 218/115 vs 1.358, 40% miss	Published
3	Elimination cascade	PHYS-14	15 candidates, 2 survivors, 1 minimal	Published
4	Cabibbo Doublet identification	PHYS-15	(3,2,1/6), gap 38/27, M GUT = 10^15.5	Published
5	Two roads converge	PHYS-16	Gap ratio path + anomaly path on same particle	Published
6	Generation democracy	PHYS-17	(4/3,4/3,4/3), fermions contribute 0% to gap	Published
7	The boson problem	PHYS-17	Gap ratio is 96-101% gauge + 0% fermion	Published
8	Y = 1/6 mechanism	PHYS-18	Db 2/Db 1 = 15, 1/Y^2 scaling law	Published
9	Three anomalies	PHYS-19	CKM deficit + A FB^b + Higgs excess	Published
10	Proton decay test	PHYS-20	tau ~ 10^34-35 yr, Hyper-K tests within decade	Published
11	Level 1 extends to BSM	PHYS-21	Unified boundary map, first unobserved particle	Published
12	A 2 geometric cancellation	PHYS-22	87% cancellation, R 4 connection	Published
13	Koide C 3 closure	PHYS-23	Tautology + saddle point, a^2 = 2 is the problem	Published

#	Finding	Paper	Content	Status
14	82/82 PSLQ null	MATH-6	Bessel zeros independent at 100 digits	Published
15	Lattice ratio independence	DATA-4	FLAG ratios independent of PDG masses, 28% discrepancy	Published
16	Two-loop improvement	(PHYS-24 pending)	Delta: -1.17 to -0.40, 66% improvement	Computed

Findings 6 and 7 share PHYS-17 because they are two aspects of the same decomposition (the generation democracy and the boson problem are discovered in the same computation). Finding 16 is computed with full supporting tables but the paper is pending.

---

4. The Papers

4.1 Published Papers

#	Registry	Title	Script	Checks
1	DATA-3	Verified Integer Rational Database Round Three	data 2 to 3 test 1.py	32/32
2	PHYS-12	Electroweak Integer Anatomy	GUT script	9/9
3	PHYS-13	The Gap Ratio	GUT script	9/9
4	PHYS-14	The Enumeration	GUT script	9/9
5	PHYS-15	The Cabibbo Doublet	GUT script	9/9
6	PHYS-16	The Two Roads	GUT script + anomaly matrix	9/9
7	PHYS-17	Generation Democracy and the Boson Problem	GUT script	9/9

#	Registry	Title	Script	Checks
8	PHYS-18	The Y = 1/6 Asymmetry	GUT script	9/9
9	PHYS-19	Independent Anomaly Evidence	Anomaly matrix (web-verified)	9/9
10	PHYS-20	The Proton Decay Test	GUT script + web search	9/9
11	PHYS-21	The Level 1 / Level 2 Boundary	All prior scripts	—
12	PHYS-22	The A 2 Geometric Cancellation	a 2 decomposition 0.py	7/7
13	PHYS-23	The Koide C 3 Closure	DATA-3 masses	32/32
14	MATH-6	The 82/82 Independence Record	bessel pslq 0.py	6/6
15	DATA-4	Verified Database with Cabibbo Doublet Extension	data 4.py	38/38
16	—	Cabibbo Doublet Database Record	GUT script + anomaly matrix	9/9

4.2 Pending Papers

#	Title	Status	Content
17	PHYS-24: Two-Loop Unification	Computation complete, paper pending	Delta: -1.17 to -0.40, supporting tables written

Every published paper has a full set of supporting appendix tables (lettered A through I depending on paper). These tables contain pre-computed data, verification checks, Level 1/Level 2 classifications, source material lists, and cross-references.

---

5. The Scripts and Verified Computations

Script	Checks	Key Output	Status
sin2 theta w 1.py	9/9	Gap ratios, BSM enumeration, M GUT	Verified
a 2 decomposition 0.py	7/7	A 2 three-piece, 87% cancellation	Verified
bessel pslq 0.py	6/6	82/82 independence, 100-digit nulls	Verified
data 2 to 3 test 1.py	32/32	DATA-3 consistency across 123 entries	Verified
data 4.py	38/38	DATA-4 consistency across 146 entries	Verified
unification test.py	6/6	Two-loop Delta = -0.40 at M VL = 500 GeV	Verified

Total verified checks across all scripts: 98.

---

6. The Parked Notebooks

Six notebooks were produced during the computational phase. Four remain parked with specific blockers and paths forward. Two were consumed by papers.

Notebook	Status	Blocker	Path Forward	Priority
sin^2 theta W from 3/8	Parked	Crossing scale undetermined	Compute L X with Cabibbo Doublet betas	MEDIUM
4-Loop Wall (A 4)	Parked	Laporta master integrals private	Wait for data OR transcribe T+V+W+E	HIGH (external)
Higgs lambda = g'^2	Parked	No derivation of impedance matching	One-loop pole mass or soliton derivation	MEDIUM-LOW
CKM from mass ratios	Parked	Precision floor + literature unknown	Web-search texture-zero literature	MEDIUM

Notebook	Status	Blocker	Path Forward	Priority
Bessel PSLQ	Consumed	None	Published as MATH-6	Done
A 2 decomposition	Consumed	None	Published as PHYS-22	Done

The sin^2 theta_W notebook is NOT definitively closed — unlike the claim in the original handoff, the two-loop unification computation from this session now provides the GUT-scale physics that was blocking the formula. Computing L_X with the Cabibbo Doublet betas and checking whether sin^2 theta_W = 0.23122 emerges is a 10-line computation for Session 4.

---

7. The Closed Paths

One investigation path was definitively closed in Session 3.

The C_3 path to Koide (PHYS-23). The 120-degree spacing is a tautology (3 parameters, 3 data points — any three positive masses fit). K = 2/3 is a saddle point of the C_3 potential, not a minimum. The path is doubly dead. The real problem — derive a^2 = 2 from physics — remains open.

The SM non-unification (PHYS-13) is an established fact, not a closed path. The sin^2 theta_W from 3/8 notebook is blocked but not closed — it has a specific computable path forward via the Cabibbo Doublet betas.

---

8. The Two-Loop Unification Extension

The unification script (unification_test.py, 6/6 checks) computed two-loop running with the SM b_ij matrix and a step-function threshold at M_VL for the Cabibbo Doublet.

Approximation	Delta(1/alpha 3) at crossing	M GUT	Quality
SM one-loop (no VL)	-6.58	10^13.8	Excluded
SM+VL one-loop	-1.17	10^15.43	Poor
SM+VL one-loop + two-loop SM b ij	-0.40	10^15.46	Near
+ VL two-loop (estimated)	~-0.27	~10^15.47	Near
+ GUT threshold corrections	~0	~10^15.5	Closable

The two-loop SM b_ij matrix (199/50, 27/10, 44/5; 9/10, 35/6, 12; 11/10, 9/2, -26) is stored in DATA-4 as entry N14. All entries are exact Fractions from Machacek-Vaughn (1983) and Luo-Xiao (hep-ph/0207271).

The dominant effect: b_33 = -26 slows SU(3) running at two loops, bringing 1/alpha_3 closer to the crossing point. The improvement from -1.17 to -0.40 is a 66% reduction in the unification miss.

The residual Delta = -0.40 is within the standard range for GUT threshold corrections in minimal SU(5) with ordinary mass splittings (factor 2-5 between heavy particles).

The Cabibbo Doublet at two loops achieves the same unification quality as the MSSM: both need GUT threshold corrections of comparable magnitude to go from “near” to “exact.”

18 supporting tables for PHYS-24 are written. The paper itself is pending.

---

9. The Running Curve Analysis

The gauge coupling running is not linear at two loops. The one-loop running equation in HOWL form:

d(1/alpha_i)/d(ln mu) = -b_i/(8R_2)

where R_2 = pi/4 is the 2D geometric modulus. The slope at each energy is an exact rational divided by 8R_2.

At two loops:

d(1/alpha_i)/d(ln mu) = -b_i/(8R_2) - sum_j b_ij alpha_j/(256R_4)

where R_4 = pi^2/32 is the 4D geometric modulus. The curvature involves R_4. The ratio of two-loop to one-loop is R_2/(32R_4) = 1/(4pi).

The running curve for alpha_3 satisfies:

1/alpha_3 - [b_33/(4pi b_3)] ln(1/alpha_3) = b_3/(8R_2) ln(mu/M_Z) + constant

This is a Lambert W function — not linear, not logarithmic, but parametrized entirely by R_2, R_4, and exact rationals from the gauge group. Each soliton boundary (mass threshold) changes the slope by Delta_b/(8R_2) and the curvature by Delta_b_ij/(256R_4).

The entire running from M_Z to M_GUT is a sequence of Lambert W segments, each with exact rational coefficients, joined at soliton boundaries. The geometric moduli R_2 and R_4 set the scale at which the integer coefficients translate into physical slopes and curvatures.

---

10. The Parameter Scorecard

Reduction	Source	Change	Status
theta QCD = 0	PHYS-7	19 to 18	Confirmed

Reduction	Source	Change	Status
m tau from Koide	PHYS-8	18 to 17	Conditional (IF a^2 = 2)
alpha s from unification	Paths 1-3	17 to 16	Not yet computed
sin^2 theta W from unification	Path 4	16 to 15	Not yet computed
M VL from threshold	Path 2	+6 - 1 net	Not yet computed
Two CKM angles from masses	Parked notebook	15 to 13	Blocked (precision floor)
lambda from impedance matching	Parked notebook	13 to 12	Blocked (no derivation)

Confirmed: 19 to 18. Conditional: 18 to 17. Computable next session: 17 to 15 (if unification works). Blocked: 15 to 12 (requires new measurements or derivations).

---

11. The Level 1 / Level 2 Inventory

From PHYS-21, updated with Session 3 results:

Level 1 results (19 entries). R_2 = pi/4, R_4 = pi^2/32, SM beta coefficients (b_1, b_2, b_3), gap ratio 218/115, generation democracy (4/3, 4/3, 4/3), boson problem decomposition, Yang-Mills coefficient 11, Higgs contribution, A_1 = 1/2, A_2 three-piece decomposition, EW integer anatomy, Cabibbo Doublet representation (3,2,1/6), Delta_b = (1/15, 1, 1/3), gap ratio 38/27, Delta_b_2/Delta_b_1 = 15, 1/Y^2 scaling, tau proportional to M_GUT^4, two-loop b_ij matrix.

Level 2 parameters (23). 17 SM parameters (after theta_QCD and conditional Koide) plus 6 Cabibbo Doublet parameters (M_VL, theta_14, theta_24, theta_34, delta_1, delta_2). All staged in DATA-4 entries 124-129.

Derived results. M_GUT = 10^15.5, tau_p ~ 10^34-35 yr, measured gap ratio 1.358, CKM deficit 0.00202, alpha to a_e at 4.3 ppb, EW observables from 7 inputs, two-loop Delta = -0.40.

---

12. The Experimental Timeline

Experiment	Observable	Prediction	Status	When
Hyper-Kamiokande	p to e+ pi0	tau ~ 10^34-35 yr (detectable)	Under construction, operations ~2027	2027-2037
HL-LHC	VL quark pair production	Observable if M VL < 2-3 TeV	Running, luminosity upgrade	Now-2040
Belle II	CKM precision (V us, V ub)	Modified first-row unitarity	Running, collecting data	Now-2030+
DUNE	Proton decay (complementary)	Detectable in some completions	Under construction	2028+

The MSSM predicts tau ~ 10^36-37 yr, beyond any planned experiment. The Cabibbo Doublet predicts tau within Hyper-K reach. This is the decisive discriminator between the two scenarios that have nearly identical gap ratios (7/5 = 1.400 vs 38/27 = 1.407).

---

13. The Database

DATA-4 is the sole data reference for all future HOWL computation. DATA-3 is retired.

Content	Count	Source
Inherited from DATA-3	123 entries	Sections A-K
Cabibbo Doublet staged	6 entries	Section L (Type G)
GUT/unification parameters	17 entries	Section N (Type D)
Total	146 entries	
Inherited checks	32	Groups A-E
New GUT checks	6	Group G (all exact)
Total checks	38	38 PASS, 0 FAIL

Finding 15 (lattice ratio independence) is formalized in DATA-4 Group F. The computation traceability map (Group T) links 12 papers to their data dependencies. When any future PDG update changes an entry, the map identifies every affected computation.

---

14. What Session 4 Should Do

Priority-ordered task list:

1. **Write PHYS-24** (two-loop unification paper). Computation complete, 18 supporting tables written.
2. **Resolve the beta normalization** (factor-of-4 discrepancy between general Machacek-Vaughn formula and verified one-loop values). Required for computing the VL two-loop b_ij.
3. **Compute VL two-loop b_ij** and rerun unification with full two-loop above M_VL.
4. **Parametrize GUT threshold corrections** in minimal SU(5) with Cabibbo Doublet. Express Delta as function of M_T/M_X, M_Sigma/M_X.
5. **Find M_VL for exact unification** (two-loop + threshold). If solution exists in 1.5-6 TeV, M_VL becomes a prediction.
6. **Compute sin^2 theta_W** from two-loop backward RGE. 10-line computation from parked notebook.
7. **Predict alpha_s** from unification condition. Consistency check: does predicted alpha_s match 0.1180?
8. **Literature search** for the mixed CKM-mass pattern (sin theta_12 = sqrt(m_d/m_s), sin theta_23 = sqrt(m_u/m_c)).
9. **Compute the running curve explicitly** (Lambert W fit, R_2/R_4 moduli at each soliton boundary).
10. **PSLQ on untested quantities** (Koide amplitudes a^2_down = 2.388, a^2_up = 3.093, amplitude ratios).

---

15. Key Numbers (Source of Truth)

All numbers from verified scripts. Any discrepancy with a paper is resolved in favor of the script output.

Quantity	Value	Source
SM gap ratio	218/115 = 1.895652	GUT script, PASS
Measured gap ratio	1.358193	GUT script from DATA-4 couplings
Cabibbo Doublet gap ratio	38/27 = 1.407407	GUT script, PASS
MSSM gap ratio	7/5 = 1.400000	GUT script, PASS
Cabibbo Doublet distance	0.049215	GUT script
M GUT (Cabibbo Doublet)	10^15.5 GeV	GUT script
M GUT (MSSM)	10^17.3 GeV	GUT script
Cabibbo Doublet Delta b	(1/15, 1, 1/3)	PHYS-15, GUT script
Delta b 2/Delta b 1	15	PHYS-18

Quantity	Value	Source
Two-loop Delta (M VL=500)	-0.40	unification test.py, PASS
One-loop Delta (M VL=500)	-1.17	unification test.py, PASS
Two-loop improvement	66%	unification test.py
A 2	-0.328478965579	A 2 script, PASS
A 2 cancellation	87.4%	A 2 script, PASS
Koide K(leptons)	0.666661	DATA-4 D1, PASS
Koide a^2(leptons)	2.0000	DATA-4 D2, PASS
Koide a^2(down)	2.3877	DATA-4 D5
Koide a^2(up)	3.0928	DATA-4 D4
Proton decay tau	~10^34-35 yr	PHYS-20
Super-K bound	tau > 2.4 x 10^34 yr	PHYS-20 (web verified)
CKM deficit	0.00202	PHYS-19
SM parameter count	17 (after theta QCD, conditional Koide)	PHYS-21
PSLQ record	82/82 null	MATH-6, PASS
DATA-4 entries	146	data 4.py, 38/38 PASS

---

16. What This Paper Does Not Claim

This paper does not claim the Cabibbo Doublet has been discovered. It has been identified by Level 1 arithmetic and corroborated by Level 2 anomalies. Discovery requires direct experimental observation.

This paper does not claim unification is proven. The gap ratio 38/27 is exact arithmetic. Whether nature chose to unify the gauge forces is a Level 2 question.

This paper does not claim all 16 findings are equally significant. Findings 2 (SM fails) and 4 (Cabibbo Doublet identified) are the load-bearing results. The others are anatomy, mechanism, tests, and infrastructure.

This paper does not claim Session 3 is complete. One paper remains pending (PHYS-24). Four parked notebooks have specific paths forward. The two-loop extension opened new computation that Session 4 should pursue.

This paper does not claim the HOWL approach is the only way to study gauge coupling unification. It is one approach — exact rational arithmetic on gauge group integers — applied systematically. The results are reproducible by anyone with the same inputs (DATA-4) and the same arithmetic.

---

17. Summary

Session 3 asked whether the Standard Model's gauge couplings unify. They do not — the gap ratio 218/115 = 1.896 misses the measured 1.358 by 40%. One particle fixes it: the Cabibbo Doublet (3,2,1/6), with gap ratio 38/27 = 1.407, identified by exact Fraction arithmetic from 15 enumerated candidates. Three independent experimental anomalies converge on the same representation from completely different data. Two-loop corrections improve unification from Delta = -1.17 to -0.40, with the residual within GUT threshold range. The proton lifetime tau ~ 10^34-35 yr is testable by Hyper-Kamiokande within a decade. The database is extended to 146 entries with 38/38 checks. The Cabibbo Doublet's quantum numbers are Level 1 — forced by the integers. Its existence is Level 2 — supplied by experiment if nature chose unification.

Sixteen findings. Fifteen papers. Six scripts. Ninety-eight checks. One particle. One question for the universe.

The integers have spoken. The universe has not yet answered.

Appendix A: Session 3 Deliverable Status (Final)

#	Paper	Status
1	DATA-3	Published
2	PHYS-12 (Electroweak Anatomy)	Published
3	PHYS-13 (Gap Ratio)	Published
4	PHYS-14 (Enumeration)	Published
5	PHYS-15 (Cabibbo Doublet)	Published
6	PHYS-16 (Two Roads)	Published
7	PHYS-17 (Generation Democracy)	Published
8	PHYS-18 (Y = 1/6 Asymmetry)	Published
9	PHYS-19 (Anomaly Evidence)	Published
10	PHYS-20 (Proton Decay Test)	Published
11	PHYS-21 (Level 1 / Level 2 Boundary)	Published
12	PHYS-22 (A 2 Geometric Cancellation)	Published
13	PHYS-23 (Koide C 3 Closure)	Published
14	MATH-6 (82/82 Independence Record)	Published

#	Paper	Status
15	DATA-4 (Verified Database Extension)	Published
16	Cabibbo Doublet Database Record	Published
17	PHYS-24 (Two-Loop Unification)	Computation complete, paper PENDING

16 of 17 deliverables published. 1 paper pending.

Appendix B: Verified Script Check Summary

Script	Group	Checks	Pass	Content
sin2 theta w 1.py	—	9	9	Normalization, gap ratios, MSSM gate, M GUT, enumeration
a 2 decomposition 0.py	—	7	7	A 2 sum, individual pieces, cancellation, Q335 basis
bessel pslq 0.py	—	6	6	j 11, j 01 references, sanity, completeness, nulls, record
data 2 to 3 test 1.py	A-E	32	32	Mass ratios, Q335, physical relations, Koide, SI exact
data 4.py	A-G	38	38	All DATA-3 checks + GUT gap ratios + modified betas
unification test.py	—	6	6	Gap ratio, normalization, SM crossing, shift, M GUT, alpha GUT
Total		**98**	**98**	

Script	Group	Checks	Pass	Content

DISC-11: Session 3 Capstone. From the gap ratio to the Cabibbo Doublet and beyond. 16 findings, 15 papers, 6 scripts, 98 checks, 1 particle, 0 contradictions. Published April 1, 2026. This paper is never edited after publication.

[**HOWL-DISC-11-2026**] From the Gap Ratio to the Cabibbo Doublet and Beyond. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HOWL-DISC-11-2026

[**HOWL-DISC-6-2026**] The Remainder Extraction Program. Github: https://github.com/ghowland/papers/tree/main/paper DISC-6-2026

[**HOWL-DISC-7-2026**] The Remainder Extraction Program: Execution and Results. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HOWL-DISC-7-2026

[**HOWL-DISC-8-2026**] The Remainder Extraction Program: Phase II Results. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HOWL-DISC-8-2026